

A NON-COMMUTATIVE ORE LCM SEMIGROUP WITH ARITHMETIC THERMODYNAMICS: THE HURWITZ QUATERNIONS

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ABSTRACT. The localized Bost–Connes programme is blocked in three directions. A commutative index semigroup makes the groupoid amenable, and Connes’ classification of injective factors then erases the arithmetic entirely (Paper IX of the series). Additive structure makes the range projections partition, so the Cuntz relation forces $\beta = 1$ and destroys the high-temperature phase structure (Paper II). A free non-commutative monoid has orthogonal ranges, whence $\sum_p \mathbf{N} p^{-\beta} \leq 1$ and no equilibrium state exists above the critical point at all (Paper V).

We isolate the algebraic conditions that avoid all three, and exhibit a monoid satisfying them. Two observations shape the requirements. First, the condition that the range projections be nested rather than partitioning is not an extra hypothesis: a right Ore semigroup has common right multiples, so in a right LCM semigroup $e_a e_b = e_{a \vee b} \neq 0$ always, and no Cuntz relation can hold. This also re-diagnoses the free monoid, whose real defect is the failure of the Ore condition rather than non-commutativity. Second, every multiplicative norm factors through the abelianization, so the available arithmetic is that of P^{ab} — which rules out any finitely generated candidate, the obstruction series then being a finite sum.

The multiplicative monoid $\mathcal{H} \setminus \{0\}$ of the Hurwitz quaternions satisfies every requirement. It is non-commutative and cancellative; right Ore, since $a(\bar{a}b) = b \text{Nrd}(a)$; right LCM, since $\mathcal{H}$ is a noncommutative principal ideal domain; its unit group has order 24, so the partition function converges somewhere, in contrast with $M_n(\mathbb{Z}) \cap GL_n^+(\mathbb{Q})$ whose unit group $GL_n(\mathbb{Z})$ is infinite; the reduced norm is multiplicative and takes every prime as a value, giving infinitely many multiplicatively independent norms; the growth is polynomial of degree two; and

$$\zeta_P(\beta) = 24 \sum_{n \geq 1} \sigma_{\text{odd}}(n) n^{-\beta} = 24 \zeta(\beta) (1 - 2^{1-\beta}) \zeta(\beta - 1),$$

a Solomon-type zeta function with a pole at $\beta = 2$: the critical exponent is finite and positive. The group of fractions $\mathbb{H}(\mathbb{Q})^\times$ is non-amenable.

We close by delimiting what remains. Non-amenableity of the group of fractions is necessary but not sufficient for escaping Paper IX: what matters there is amenability of the *measured* groupoid, and boundary actions of non-amenable groups are routinely amenable, $L^\infty(\partial \mathbb{F}_n) \rtimes \mathbb{F}_n$ being injective. We state this as a fourth condition (C4) and leave it open; in our view it is the question on which a genuinely non-commutative Bost–Connes theory now turns. Condition (C4) is settled, in the negative, in Paper XXVIII of the series: the orbit relation is hyperfinite for every admissible monoid, so the search closed there.

1. THE ALGEBRAIC REQUIREMENTS

The series. The foundational paper is [8] and Paper I is [9]; Papers II, V, VII, IX and XXVII, cited by numeral, are in preparation, and nothing proved here depends on them.

Definition 1.1 (The target). We seek a monoid P with:

- (C1) P non-commutative, cancellative, right Ore, $aP \cap bP \neq \emptyset$ for all a, b — so that a group of right fractions $G = PP^{-1}$ exists — and right LCM, meaning $aP \cap bP$ is empty or equal to cP for some c [6, 4];

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/X-semigroup-search/](#).

- (C2) in the Toeplitz representation $v_a^* v_a = 1$, $e_a = v_a v_a^*$, the range projections nested, $e_a e_b = e_{a \vee b}$, with no finite family satisfying $\sum_i e_{a_i} = 1$;
 (C3) a multiplicative $N : P \rightarrow \mathbb{R}_{>0}$ for which $\zeta_P(\beta) = \sum_{a \in P} N(a)^{-\beta}$ has a finite abscissa of convergence $\beta_c \in (0, \infty)$.

Proposition 1.2 ((C2) follows from (C1)). *Let P satisfy (C1). Then $a \vee b$ exists for all $a, b \in P$, the projections satisfy $e_a e_b = e_{a \vee b} \neq 0$, and no relation $\sum_i e_{a_i} = 1$ with orthogonal e_{a_i} and at least two nonzero terms can hold. In particular the Toeplitz algebra of P admits no Cuntz relation.*

Proof. The Ore condition gives $aP \cap bP \neq \emptyset$ for all a, b ; the LCM condition then gives $aP \cap bP = (a \vee b)P$. In the Nica–Toeplitz representation on $\ell^2(P)$ the projection e_a has range $\overline{\text{span}}\{\delta_x : x \in aP\}$, so $e_a e_b$ is the projection onto $aP \cap bP = (a \vee b)P$, namely $e_{a \vee b} \neq 0$. Orthogonality would require $aP \cap bP = \emptyset$, contradicting Ore. $\square$

Remark 1.3 (Re-diagnosis of the free monoid). Proposition 1.2 shows that (C2) costs nothing once (C1) is imposed, and identifies what actually goes wrong in the free variant of Paper V. A free monoid is right LCM in the quasi-lattice sense [6] but is *not* Ore: two distinct generators have no common right multiple. That failure is what makes the ranges orthogonal, and orthogonality is what forces $\sum_p \text{Np}^{-\beta} \leq 1$ and empties the high-temperature region. The obstruction is the failure of the Ore condition, not non-commutativity as such.

Proposition 1.4 (Norms factor through the abelianization). *Any multiplicative homomorphism $N : P \rightarrow (\mathbb{R}_{>0}, \times)$ factors through the abelianization P^{ab} . Hence the norms available to (C3) are exactly the elements of $\text{Hom}(P^{\text{ab}}, \mathbb{R}_{>0})$.*

Proof. $\mathbb{R}_{>0}$ is abelian, so N annihilates all commutators. $\square$

Remark 1.5 (What (C1)–(C3) already exclude). Two consequences narrow the search sharply, and we record them because they explain why the candidate of §2 is arithmetic rather than combinatorial. If P is finitely generated then the obstruction series of [8, (1.1)], which runs over generators, is a finite sum, convergent for every β ; so Ξ_β is constant in β and no phase transition occurs. And if the unit group $P^\times$ is infinite then, N being 1 on units, $\zeta_P(\beta) \geq |P^\times| = \infty$ for every β and (C3) fails. A candidate must therefore be infinitely generated with finite unit group, and by Proposition 1.4 its abelianization must carry infinitely many multiplicatively independent norm values — the analogue of the $\mathbb{Q}$ -linear independence of the $\log \text{Np}$ used throughout [9].

2. THE HURWITZ QUATERNION MONOID

Let $\mathbb{H}(\mathbb{Q})$ be the rational quaternion algebra, $\bar{a}$ the conjugate, $\text{Nrd}(a) = a\bar{a} \in \mathbb{Z}_{\geq 0}$ the reduced norm, and

$$\mathcal{H} = \mathbb{Z} \left\langle 1, i, j, \frac{1+i+j+k}{2} \right\rangle$$

the Hurwitz order [3, 7]. Put $P := \mathcal{H} \setminus \{0\}$ under multiplication.

Theorem 2.1 (P satisfies (C1)–(C3)). (1) P is non-commutative and cancellative, being a subsemigroup of the multiplicative group of a division algebra.

(2) P is right Ore: for $a, b \in P$,

$$a(\bar{a}b) = (a\bar{a})b = \text{Nrd}(a)b = b\text{Nrd}(a),$$

since $\text{Nrd}(a)$ is a central integer; so $aP \cap bP \neq \emptyset$.

(3) P is right LCM: $\mathcal{H}$ is Euclidean for Nrd , hence a noncommutative principal ideal domain, so every right ideal is principal and $a\mathcal{H} \cap b\mathcal{H} = c\mathcal{H}$.

(4) $P^\times$ is the group of 24 Hurwitz units, in particular finite.

(5) Nrd is multiplicative with $\text{Nrd}|_{P^\times} = 1$; its values on irreducibles are exactly the rational primes, whose logarithms are $\mathbb{Q}$ -linearly independent.

$$(6) \text{ Writing } a_n = \#\{a \in P : \text{Nrd}(a) = n\} = 24\sigma_{\text{odd}}(n),$$

$$(2.1) \quad \zeta_P(\beta) = 24 \sum_{n \geq 1} \sigma_{\text{odd}}(n) n^{-\beta} = 24 \zeta(\beta) (1 - 2^{1-\beta}) \zeta(\beta - 1),$$

convergent for $\beta > 2$ and divergent as $\beta \downarrow 2$; the critical exponent is $\beta_c = 2$.

(7) The growth is polynomial of degree two: $\#\{a \in P : \text{Nrd}(a) \leq x\} \sim Cx^2$.

Proof. (1) is immediate. (2) is the displayed identity, $\text{Nrd}(a)$ lying in the centre $\mathbb{Q}$ of $\mathbb{H}(\mathbb{Q})$ and in $\mathbb{Z}$ for $a \in \mathcal{H}$. (3) is the classical fact that $\mathcal{H}$ is left and right Euclidean for the reduced norm [3], whence a principal ideal domain on both sides; an intersection of right ideals is a right ideal, hence principal, and nonzero by (2). (4) is the classical count of Hurwitz units, namely $\pm 1, \pm i, \pm j, \pm k$ and the sixteen $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$. (5): multiplicativity is $\text{Nrd}(ab) = \text{Nrd}(a) \text{Nrd}(b)$; an element is irreducible exactly when its reduced norm is prime, and every prime occurs; the $\mathbb{Q}$ -linear independence of $\{\log p\}$ is unique factorization in $\mathbb{Z}$.

For (6), the count $a_n = 24\sigma_{\text{odd}}(n)$ is the Hurwitz–Jacobi formula [3]. Its Dirichlet series is

$$\sum_{n \geq 1} \sigma_{\text{odd}}(n) n^{-s} = \sum_{d \text{ odd}} \sum_{m \geq 1} d(dm)^{-s} = \left(\sum_{d \text{ odd}} d^{1-s} \right) \zeta(s) = (1 - 2^{1-s}) \zeta(s-1) \zeta(s),$$

whose only pole in $\Re s > 1$ is that of $\zeta(s-1)$ at $s = 2$. (7) follows by Abel summation from $\sum_{n \leq x} \sigma_{\text{odd}}(n) \asymp x^2$. $\square$

Remark 2.2 (Numerical verification). The counts a_n were obtained by exhaustive enumeration of $\mathcal{H}$ over both the integral and the half-integral coordinate classes, and agree with $24\sigma_{\text{odd}}(n)$ for every $n \leq 15$:

$$24, 24, 96, 24, 144, 96, 192, 24, 312, 144, 288, 96, 336, 192, 576.$$

Identity (2.1) was checked against partial sums to $n = 20000$, agreeing to eight decimal places at $\beta = 4$ and $\beta = 6$. The pole at $\beta = 2$ is visible in

$$\zeta_P(2.20) = 112.96, \quad \zeta_P(2.10) = 211.44, \quad \zeta_P(2.05) = 408.72, \quad \zeta_P(2.02) = 1000.83.$$

The ratio $\#\{\text{Nrd} \leq x\}/x^2$ took the values 9.868, 9.880, 9.872 at $x = 200, 500, 1000$, confirming (7).

Remark 2.3 (The Toeplitz relations). The associated algebra is the Nica–Toeplitz algebra $\mathcal{T}(P)$ [6, 4], generated by isometries v_a , $a \in P$, subject to

$$v_a v_b = v_{ab}, \quad v_a^* v_a = 1, \quad (v_a v_a^*)(v_b v_b^*) = v_{a \vee b} v_{a \vee b}^*,$$

the third by Proposition 1.2, together with unitarity of v_u for the 24 units. The dynamics is $\sigma_t(v_a) = \text{Nrd}(a)^{it} v_a$. This is well defined and is a one-parameter group of $*$ -automorphisms preserving all three relations: multiplicativity of Nrd gives $\sigma_t(v_a v_b) = \sigma_t(v_{ab})$; and since $\text{Nrd}(u) = 1$ for each of the 24 units,

$$\sigma_t(v_u) = v_u \quad (u \in P^\times, t \in \mathbb{R}),$$

so σ_t fixes the unitaries v_u pointwise. In particular σ_t is constant on each principal right ideal, a and au having equal reduced norm, which is what makes the dynamics descend to the projections e_a indexed by $P/P^\times$. By Proposition 1.2 no Cuntz relation constrains β : the high-temperature region $\beta < 2$ is not excluded on those grounds, in contrast with both the affine systems of Paper II and the free variant of Paper V. The ring C^* -algebra of [2] is the algebra of the affine kind, with the additive structure that forces the Cuntz relation, and is not the object here.

Proposition 2.4 (The group of fractions is non-amenable). $G = PP^{-1} = \mathbb{H}(\mathbb{Q})^\times$ is non-amenable.

Proof. Every element of $\mathbb{H}(\mathbb{Q})^\times$ is ab^{-1} with $a \in \mathcal{H}$ and $b \in \mathbb{Z} \setminus \{0\}$, so $G = \mathbb{H}(\mathbb{Q})^\times$. Conjugation $x \mapsto axa^{-1}$ on the pure quaternions embeds $\mathbb{H}(\mathbb{Q})^\times/\mathbb{Q}^\times$ into $SO(3)$. The quaternions of reduced norm 5 with odd positive real part and even imaginary parts are $1 \pm 2i$, $1 \pm 2j$, $1 \pm 2k$, and by Lubotzky–Phillips–Sarnak [5, §3] the elements $1 + 2i$, $1 + 2j$, $1 + 2k$ generate a free group of rank three modulo the centre — their images in $SO(3)$ are the rotations by $\arccos(-\frac{3}{5})$ about the three coordinate axes. So $\mathbb{H}(\mathbb{Q})^\times$ contains a free subgroup of rank two and is not amenable. $\square$

Remark 2.5 (Multiplicity at each prime). Unlike the classical setting, where each rational prime contributes a single generator, here each odd prime p carries $p + 1$ distinct principal right ideals of reduced norm p , since $a_p/|P^\times| = 24(1 + p)/24 = p + 1$ by Theorem 2.1(6).

This count is geometric. For p odd the algebra is split, $\mathbb{H} \otimes \mathbb{Q}_p \cong M_2(\mathbb{Q}_p)$, and the $p + 1$ ideals are the $p + 1$ neighbours of a vertex in the Bruhat–Tits tree of $SL_2(\mathbb{Q}_p)$, a $(p + 1)$ -regular tree, indexed by $\mathbb{P}^1(\mathbb{F}_p)$. At $p = 2$, where $\mathbb{H}$ is ramified, the count collapses to a single ideal — consistent with the factor $(1 - 2^{1-\beta})$ in (2.1), and verified numerically alongside $p + 1$ for $p = 3, 5, 7, 11, 13, 17, 19, 23$.

Two consequences. The Kakutani analysis of [8, §3] was carried out with the generators indexed by primes, one per norm value; whether the multiplicity $p + 1$ alters it we have not examined, and it is the first thing to check before transporting the classification. And the tree picture suggests the shape a replacement would take: the obstruction series would be indexed not by primes but by edges, so the criterion should become a statement about a random walk on the product of the local trees rather than about a Dirichlet series in $\mathbb{N}p^{-\beta}$. We record this as an interface, not a result.

Remark 2.6 (Ramanujan graphs). The same $(p + 1)$ -regular trees, quotiented by congruence subgroups of the unit group of an order in $\mathbb{H}$, are the source of the Lubotzky–Phillips–Sarnak Ramanujan graphs [5]. We have not used this and make no claim about a connection; we note only that the arithmetic input here — the Hurwitz order and its ideals of prime norm — is the same input, and that any spectral estimate needed for (C4) would plausibly meet it.

3. THE REMAINING CONDITION

Remark 3.1 (What decided the collapse in Paper IX). Conditions (C1)–(C3) are necessary but not sufficient, and they do not address the property that governed Paper IX. There the factor was erased not because the index semigroup was commutative as such, but because the *measured* groupoid $(G \ltimes \Omega, \nu)$ was amenable, so that Connes’ classification of injective factors left only the type. Proposition 2.4 supplies non-amenability of G , which is necessary; it is far from sufficient. Boundary actions of non-amenable groups are routinely amenable in Zimmer’s sense: the action of a free group $\mathbb{F}_n$ on its Gromov boundary with any quasi-invariant measure is amenable, so $L^\infty(\partial\mathbb{F}_n) \rtimes \mathbb{F}_n$ is an injective factor. If the KMS_β measures of the Hurwitz system concentrate on a boundary of that kind, the measured groupoid is amenable, the factor is injective, and Connes’ classification applies verbatim as in Paper IX, collapsing the factor to R_∞ , R_λ or a Krieger factor.

Definition 3.2 (The fourth condition). (C4) For β in the symmetry-breaking regime the measured groupoid $(G \ltimes \Omega, \nu_\beta)$ is non-amenable; equivalently, by Anantharaman–Delaroche and Renault [1], $\pi_\varphi(\mathcal{T}(P))''$ is not injective.

Remark 3.3 (Status, and how it might go). We have not decided (C4), and we regard it as the substance of the problem rather than a technical residue. Two considerations pull in opposite directions. For: the space Ω of a Toeplitz algebra is not a boundary. The nested projections of Proposition 1.2 mean precisely that Ω retains an interior, which is the geometric content of the absence of a Cuntz relation, so the analogy with $\partial\mathbb{F}_n$ is not immediate. Against: for β below the critical exponent the KMS_β measures do concentrate away from that interior, which is where the boundary analogy would bite.

4. ASSESSMENT

Condition	Status for $P = \mathcal{H} \setminus \{0\}$	Reference
non-commutative, cancellative	yes	Thm. 2.1(1)
right Ore	yes: $a(\bar{a}b) = b \text{Nrd}(a)$	Thm. 2.1(2)
right LCM	yes: $\mathcal{H}$ a noncommutative PID	Thm. 2.1(3)
(C2) nested, no Cuntz relation	automatic from (C1)	Prop. 1.2
$ P^\times < \infty$	yes, $= 24$	Thm. 2.1(4)
infinitely many independent norms	yes: all primes	Thm. 2.1(5)
(C3) finite β_c	yes, $\beta_c = 2$	Thm. 2.1(6)
sub-exponential growth	yes: degree 2	Thm. 2.1(7)
G non-amenable	yes	Prop. 2.4
(C4) measured groupoid non-amenable	open	Def. 3.2

The Hurwitz order supplies a non-commutative, right Ore, right LCM monoid with a genuinely arithmetic multiplicative norm, a Solomon-type partition function with a finite critical exponent, polynomial growth, and a non-amenable group of fractions. Every algebraic obstacle that closed the earlier routes is absent: there is no Cuntz relation to force $\beta = 1$, no orthogonality to empty the high-temperature region, and the index semigroup is not commutative.

Three questions remain, in order of importance. First (C4), on which everything turns. Second, the multiplicity of Remark 2.5: the $p + 1$ right ideals of norm p have no counterpart in [8], and the Kakutani criterion must be re-derived before the classification can be transported. Third, if (C4) holds, whether the resulting obstruction group is genuinely non-abelian, or whether the mechanism of Paper VII — that the image of a commutative index set in the symmetry group is abelian — reappears in some form despite the non-commutativity of P ; Paper XXVII shows that non-abelian symmetry breaking to a normal subgroup can already be produced with the commutative J_S through a non-abelian cocycle, at the price of a dependence on the lifts of the Frobenius classes, so what a non-commutative P would have to add is canonicity.

We do not claim novelty for the Hurwitz order as an object of study; Bost–Connes and Hecke-algebra constructions attached to orders in quaternion algebras have been considered, and (2.1) is a classical Solomon-type zeta function. What is claimed is the selection: that the requirements of §1 admit this model, with the properties computed in §2. Whether the KMS classification of $\mathcal{T}(\mathcal{H} \setminus \{0\})$ is already known we have not been able to verify, and it should be checked before any priority claim is made.

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